

Statistics 6910, Section 003, Fall 2013, Quiz 1 (25 Points)

Wednesday, September 11, 2013

Your Name: _____

Question 1: Data (8 Points)

We have discussed several different types of data, other than the typical numerical (and categorical) data we have used in most of the other statistics classes. List one additional type of data a statistician may have to analyze: _____

Outline a scenario where such data may be encountered and formulate one possible research question a statistician might be asked to answer for such data:

Think of a possible approach *how* this question can be answered in a statistical way and outline your approach in 2 or 3 sentences:

Question 2: Uniform Distribution (9 Points)

Similarly to the creation of random numbers (RNs) from a normal distribution, we can create RNs from many other distributions. Here, we are interested in RNs from a uniform distribution. Below is the relevant information from the help page:

The Uniform Distribution

Description

These functions provide information about the uniform distribution on the interval from `min` to `max`. `dunif` gives the density, `punif` gives the distribution function, `qunif` gives the quantile function and `runif` generates random deviates.

Usage

```
[...]  
runif(n, min = 0, max = 1)
```

Arguments

```
[...]  
n  
number of observations. If length(n) > 1, the length is taken to be the  
number required.
```

`min, max`

lower and upper limits of the distribution. Must be finite.

Suppose we want to generate 10 random values from a uniform distribution with $min = 0$ and $max = 10$. Assume that we execute these expressions at the start of an R session with no previous workspace loaded. Which of the following commands is suitable for this task? **Circle the correct one(s)**. We start with 9 points. For each missing correct answer, you will lose points — and for each additional incorrect answer, you will also lose points. Not marking anything at all will result in 0 points.

`runif(0, 10)`

`runif(0, 10, 10)`

`runif(10, 0, 10)`

`runif(10, 10, 0)`

`runif(n = 10, max = 10, min = 0)`

`runif(n = 10, max = n)`

`runif(10, max = 10)`

Question 3: Variables and Assignments (8 Points)

We want to assign the value -5 to a variable. Which of the following expressions will do this? Think of valid variable names, assignment operators, and breaks. Read carefully! Assume that we sequentially (!) execute these expressions at the start of an R session with no previous workspace loaded. For each of the 8 expressions (sometimes more than just 1 line) mark the best possible answer with an [x] **above** the expression.

```
# 1 The following expression will
#      [ ] make the correct assignment
#      [ ] do something (but not the correct assignment)
#      [ ] result in an error
```

```
x1y10 = - 5
```

```
# 2 The following expression will
#      [ ] make the correct assignment
#      [ ] do something (but not the correct assignment)
#      [ ] result in an error
```

```
x1y10 == -5
```

```
# 3 The following expression will
#      [ ] make the correct assignment
#      [ ] do something (but not the correct assignment)
#      [ ] result in an error
```

```
abc99 < - 5
```

```
# 4 The following expression will
#      [ ] make the correct assignment
#      [ ] do something (but not the correct assignment)
#      [ ] result in an error
```

```
_myX = -5
```

```
# 5 The following expression will
#      [ ] make the correct assignment
#      [ ] do something (but not the correct assignment)
#      [ ] result in an error
```

```
x99_y100 = -5
```

```
# 6 The following expression will
#      [ ] make the correct assignment
#      [ ] do something (but not the correct assignment)
#      [ ] result in an error
```

```
x50_999<--5
```

```
# 7 The following expression will
#      [ ] make the correct assignment
#      [ ] do something (but not the correct assignment)
#      [ ] result in an error
```

```
xyz =
-5
```

```
# 8 The following expression will
#      [ ] make the correct assignment
#      [ ] do something (but not the correct assignment)
#      [ ] result in an error
```

```
jjj <- -
5
```